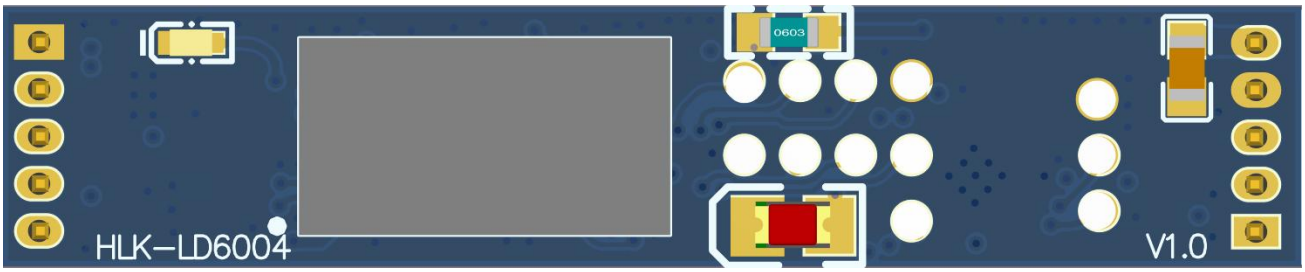
Shenzhen Hi-Link Electronic Co.,Ltd

HLK-LD6004

3D Presence Radar Module

Communication Protocol



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1. Purpose

This document defines the communication protocol for the radar module.

2. Range

This document is intended for project developers, maintenance personnel, testers, and clients.

3. Data communication link agreement

3.1. Introduction to communication link

The server refers to the module side, and the client refers to the host computer side.

Server-side communication methods: active upload and passive upload.

Client communication method: Proactive sending.

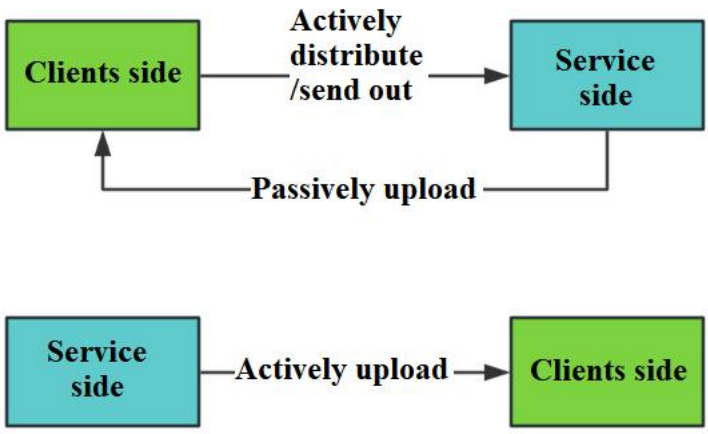


Figure 1: Communication methods

3.2. Example of communication information

Message type: Check firmware status (TYPE: 0xFFFF)

Communication method	Proactively distribute/send out		
Format	Bytes	Basic types	Frame structure
SOF	1 byte	uint8	Starting frame
ID	2 byte	uint16	Frame ID
LEN	2 byte	uint16	Data frame length
TYPE	2 byte	uint16	Frame type
HEAD_CKSUM	1 byte	uint8	Header checksum

Example TF frame: 01 00 01 00 00 FF FF FF

Message type: Return firmware status (TYPE: 0xFFFF)

Communication method	Passively upload		
Format	Bytes	Basic types	Frame structure
SOF	1 byte	uint8	Starting frame
ID	2 byte	uint16	Frame ID
LEN	2 byte	uint16	Data frame length
TYPE	2 byte	uint16	Frame type
HEAD_CKSUM	1 byte	uint8	Header checksum
DATA (see below)			
DATA_CKSUM	1 byte	uint8	Data checksum

The following describes the meaning of the DATA bits:

DATA			
Bytes	Basic types	Frame structure	Frame meaning
1 byte	uint8	[project_name]	Indicates the project currently running on the radar: 0: Indicates a general sensing project. 1: Indicates a respiration detection project. 2: Indicates a gesture detection project. 3: Indicates a distance measurement project. 4: Indicates a people counting project. 5: Indicates a 3D point cloud detection project.
1 byte	uint8	[major_version]	Indicates the radar's main version number.
1 byte	uint8	[sub_version]	Indicates the radar sub-version number.
1 byte	uint8	[modified_version]	Indicates the radar revision number.

Example TF frame: 01 04 AF 00 04 FF FF 51 06 03 0A 00 F0

Message type: Upload target information (TYPE: 0x0A08)

Communication method	Actively upload		
Format	Bytes	Basic types	Frame structure
SOF	1 byte	uint8	Starting frame
ID	2 byte	uint16	Frame ID
LEN	2 byte	uint16	Data frame length
TYPE	2 byte	uint16	Frame type
HEAD_CKSUM	1 byte	uint8	Header checksum
DATA (see below)			
DATA_CKSUM	1 byte	uint8	Data checksum

The following describes the meaning of the DATA bits:

DATA			
Bytes	Basic types	Frame structure	Frame meaning
4 byte	int32	[target_num]	Number of targets
4 byte	float	[x]	x-coordinate, data type: float, unit: m
4 byte	float	[y]	y-coordinate, data type: float, unit: m
4 byte	float	[z]	z-coordinate, data type: float, unit: m
4 byte	int32	[dop_idx]	Data type: int32, Speed: dop_idx
4 byte	Int32	[cluster_id]	Data type: int32, Clustering target ID
…	…	…	
Note: When there are N targets, x, y, z, dop_idx, and cluster_id will also have N values.			

Example TF frame: 01 00 06 00 04 0A 08 FE 00 00 00 00 FF

4. Agreement terms and conditions

4.1. Protocol structure block diagram

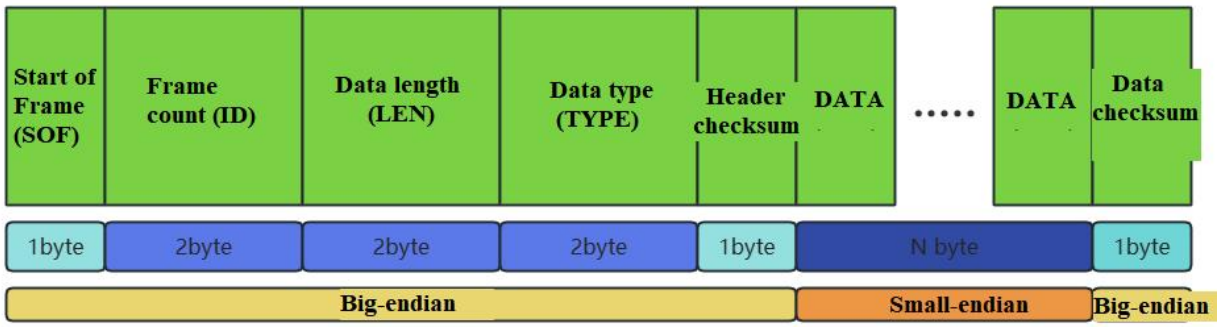


Figure 2: Protocol structure

4.2. Introduction to protocol structure

Format	Bytes	Storage method	Format meaning
SOF	1	Big-endian	Represents the starting byte of a TF data frame, usually fixed at 0x01.
ID	2	Big-endian	This represents the packet sequence number being sent, which can be used to detect whether the data frames between the client and server are continuous.
LEN	2	Big-endian	The program's default settings limit the total data frame length to no more than 1024 bytes.
TYPE	2	Big-endian	This represents the message type for the entire data frame; different message types represent different functions.

HEAD_CKSUM	1	Big-endian	This represents the header frame checksum, which is calculated by performing a bitwise XOR operation on all bits from the SOF bit to the TYPE bit, and then inverting the result.
DATA	N	Small-endian	The data payload represents the actual data; the length of the DATA bits is related to the LEN bits. For example, if the value of the LEN bits is 2, then the DATA bits consist of two bytes of data.
DATA_CKSUM	1	Big-endian	This represents the trailer frame checksum, which is calculated by performing a bitwise XOR operation on all DATA bits, followed by a bitwise NOT operation.

Note: The reason the SOF bit to HEAD_CKSUM bit and DATA_CKSUM use big-endian ordering is to facilitate viewing the instruction's TYPE and data length. The DATA bits use little-endian ordering, which conforms to the computer's storage order and facilitates parsing the DATA data (allowing for forced data type conversion).

4.3. TF frame parsing process

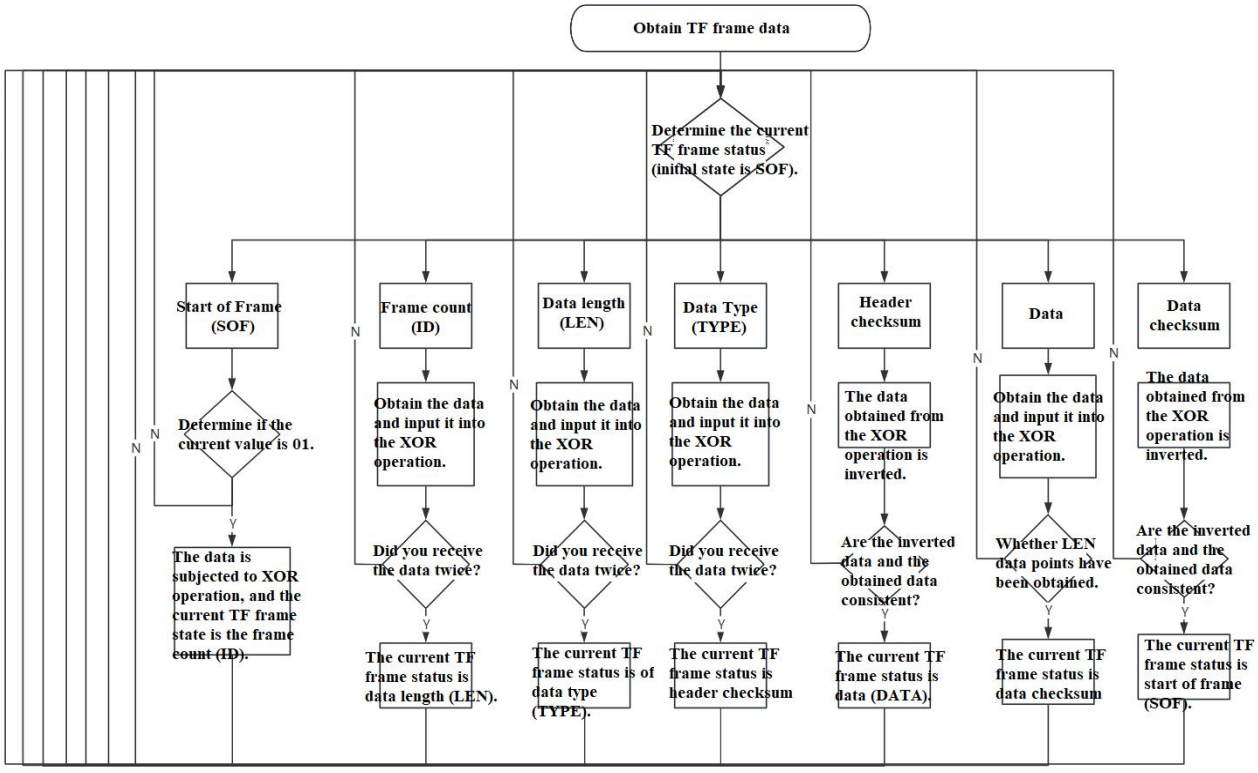
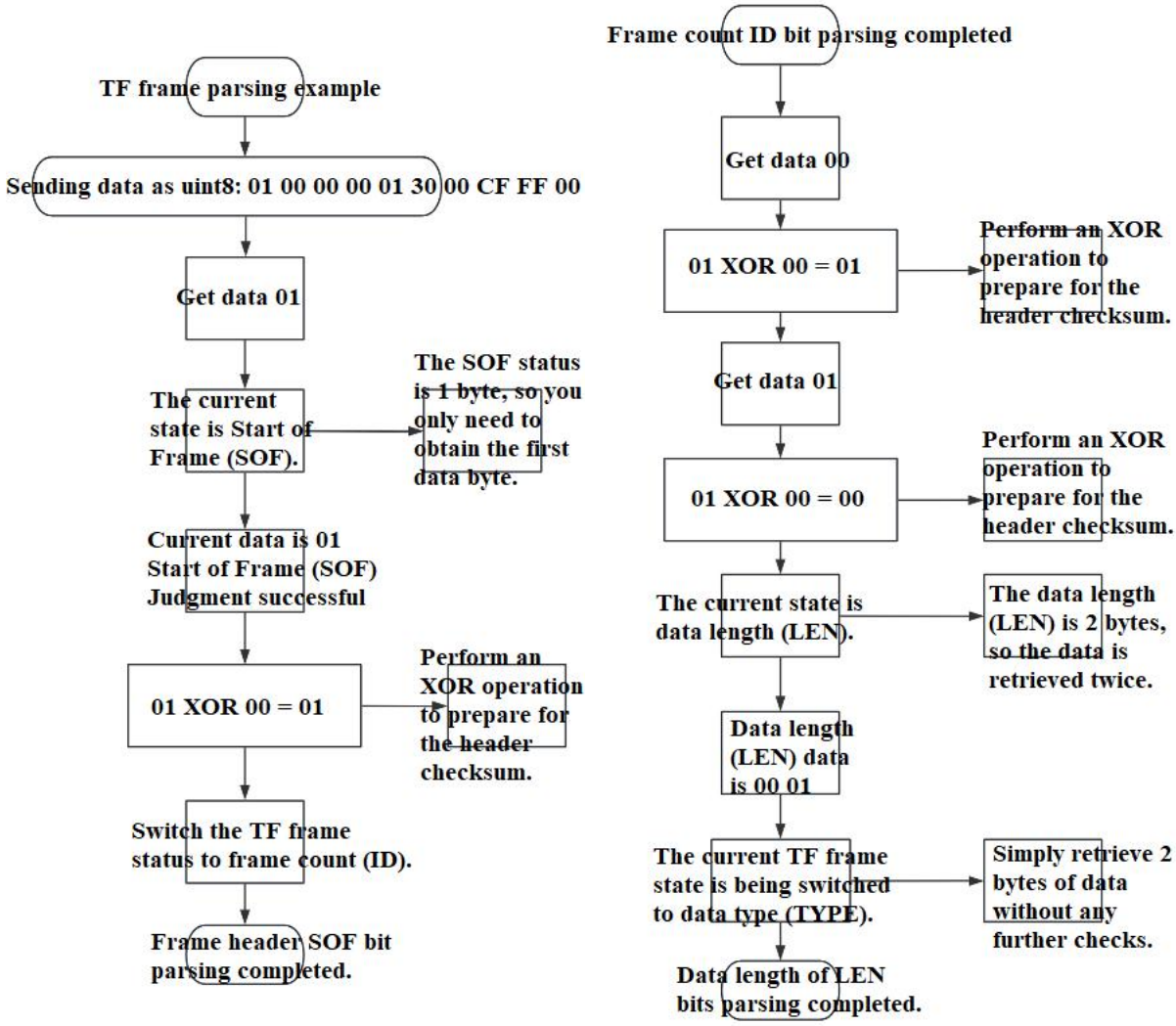
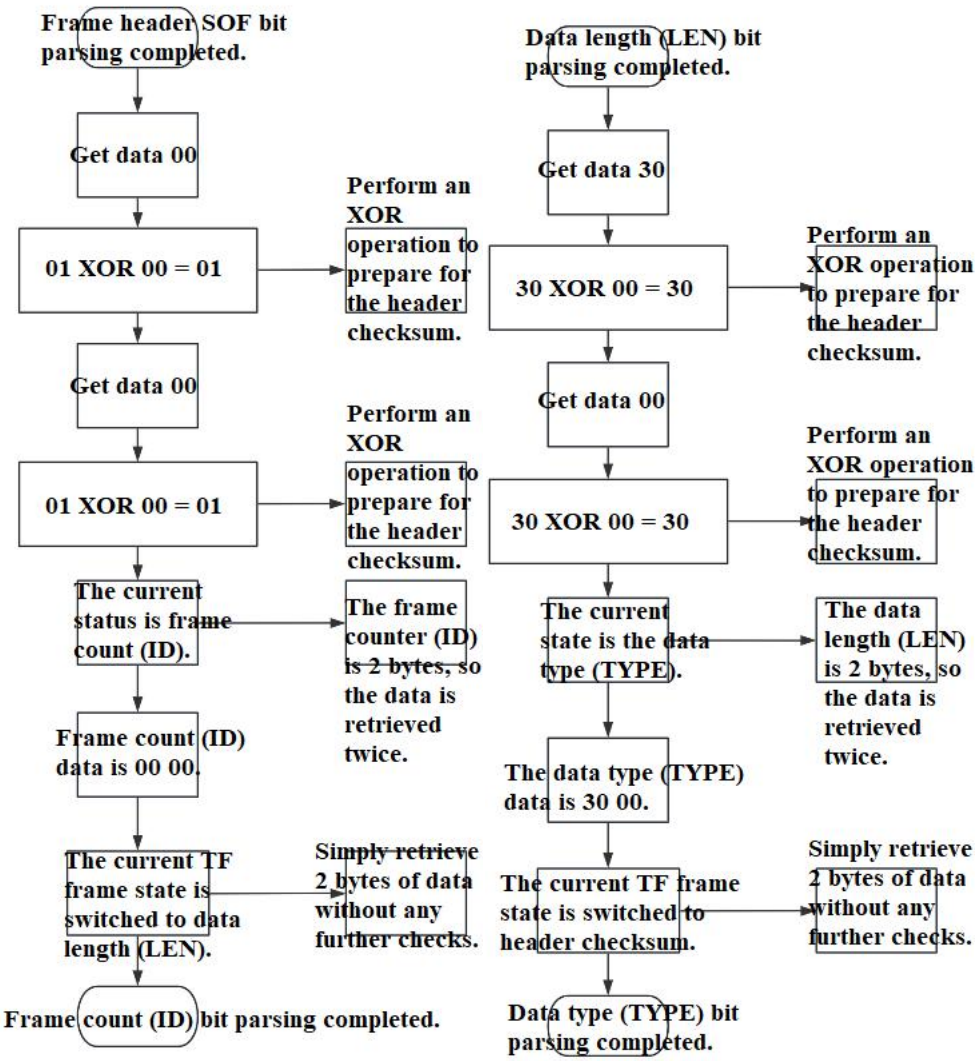
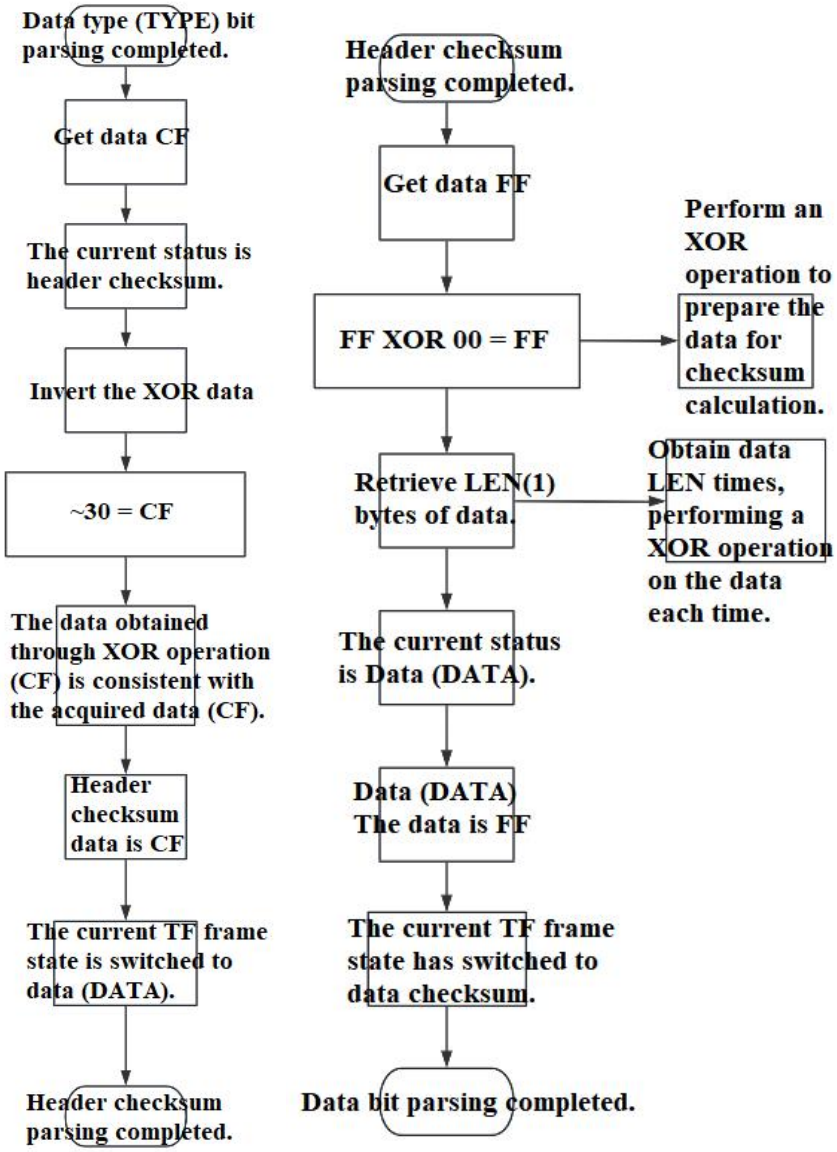


Figure 3: TF frame parsing process

4.4. Detailed explanation of TF example frame parsing




Frame header SOF bit parsing completed.
Get data 00
01 XOR 00 = 01
Perform an XOR operation to prepare for the header checksum.
Get data 00
01 XOR 00 = 01
Perform an XOR operation to prepare for the header checksum.
The current status is frame count (ID).
The frame counter (ID) is 2 bytes, so the data is retrieved twice.
Frame count (ID) data is 00 00.
The current TF frame state is switched to data length (LEN).
Simply retrieve 2 bytes of data without any further checks.
Frame count (ID) bit parsing completed.
Data length (LEN) bit parsing completed.
Get data 30
30 XOR 00 = 30
Perform an XOR operation to prepare for the header checksum.
Get data 00
30 XOR 00 = 30
Perform an XOR operation to prepare for the header checksum.
The current state is the data type (TYPE).
The data length (LEN) is 2 bytes, so the data is retrieved twice.
The data type (TYPE) data is 30 00.
The current TF frame state is switched to header checksum.
Simply retrieve 2 bytes of data without any further checks.
Data type (TYPE) bit parsing completed.


Data type (TYPE) bit parsing completed.
Get data CF
The current status is header checksum.
Invert the XOR data
~30 = CF
The data obtained through XOR operation (CF) is consistent with the acquired data (CF).
Header checksum data is CF
The current TF frame state is switched to data (DATA).
Header checksum parsing completed.
Header checksum parsing completed.
Get data FF
FF XOR 00 = FF
Perform an XOR operation to prepare the data for checksum calculation.
Retrieve LEN(1) bytes of data.
Obtain data LEN times, performing a XOR operation on the data each time.
The current status is Data (DATA).
Data (DATA) The data is FF
The current TF frame state has switched to data checksum.
Data bit parsing completed.

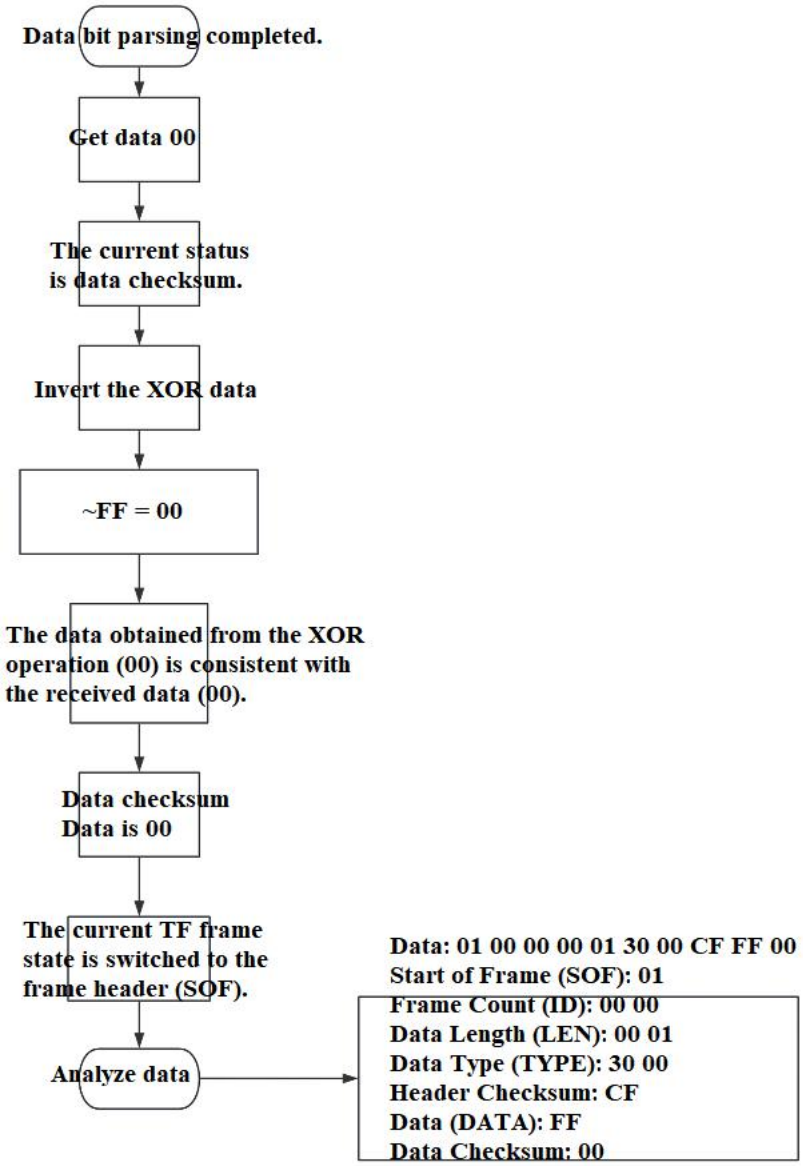


Figure 4: Detailed explanation of TF example frame parsing

5. Generic message type

5.1. Message type: Check firmware status (TYPE: 0xFFFF)

Communication method	Proactively distribute/send out		
Format	Bytes	Basic types	Frame structure
SOF	1 byte	uint8	Starting frame
ID	2 byte	uint16	Frame ID
LEN	2 byte	uint16	Data frame length
TYPE	2 byte	uint16	Frame type
HEAD_CKSUM	1 byte	uint8	Header checksum

Example TF frame: 01 00 01 00 00 FF FF FF

5.2. Message type: Return firmware status (TYPE: 0xFFFF)

Communication method	Passively upload		
Format	Bytes	Basic types	Frame structure
SOF	1 byte	uint8	Starting frame
ID	2 byte	uint16	Frame ID
LEN	2 byte	uint16	Data frame length
TYPE	2 byte	uint16	Frame type
HEAD_CKSUM	1 byte	uint8	Header checksum
DATA (see below)			
DATA_CKSUM	1 byte	uint8	Data checksum

The following explains the meaning of the DATA bits:

DATA			
Bytes	Basic types	Frame structure	Frame meaning
1 byte	uint8	[project_name]	Indicates the project currently running on the radar: 0: Indicates a general sensing project. 1: Indicates a respiration detection project. 2: Indicates a gesture detection project. 3: Indicates a distance measurement project. 4: Indicates a people counting project. 5: Indicates a 3D point cloud detection project.
1 byte	uint8	[major_version]	Indicates the radar's main version number.
1 byte	uint8	[sub_version]	Indicates the radar sub-version number.
1 byte	uint8	[modified_version]	Indicates the radar revision number.

Example TF frame: 01 04 AF 00 04 FF FF 51 06 03 0A 00 F0

5.3. Message type: Entering OTA upgrade (TYPE: 0x3000)

Communication method	Proactively distribute/send out		
Format	Bytes	Basic types	Frame structure
SOF	1 byte	uint8	Starting frame
ID	2 byte	uint16	Frame ID
LEN	2 byte	uint16	Data frame length
TYPE	2 byte	uint16	Frame type
HEAD_CKSUM	1 byte	uint8	Header checksum

Example TF frame: 01 00 01 00 00 30 00 CF

5.4. Message Type: OTA upgrade (TYPE: 0x3000)

Communication method	Passively upload		
Format	Bytes	Basic types	Frame structure
SOF	1 byte	uint8	Starting frame
ID	2 byte	uint16	Frame ID
LEN	2 byte	uint16	Data frame length
TYPE	2 byte	uint16	Frame type
HEAD_CKSUM	1 byte	uint8	Header checksum
DATA (see below)			
DATA_CKSUM	1 byte	uint8	Data checksum

The following explains the meaning of the DATA bits:

DATA			
Bytes	Basic types	Frame structure	Frame meaning
1 byte	uint8	[tmpCodeInfo]	FF: To be upgraded

Example TF frame: 01 00 03 00 01 30 00 CC FF 00

6. Presence sensing 2D/3D projects

Special instructions:

1. In this project, for all TF frames sent to the lower-level device, the lower-level device will first reply with a message of the same TYPE but without the DATA field to inform the upper-level device that it has received the data. If there is no reply, please resend the configuration message.

2. When the low-power mode for unattended operation is enabled, the slave device will enter sleep mode. Before sending configuration messages, the RX0 pin of UART0 needs to be pulled low to wake it up. Alternatively, the slave device can be woken up by sending a configuration message, and the response can be read back to confirm whether the message was configured successfully. If there is no response, the configuration message should be resent.

3. After the host sends data, or the slave device is woken up, the slave device will operate in normal mode and will not enter the low-power mode for unattended operation for 10 seconds. If no activity is detected after 10 seconds, it will enter the low-power mode for unattended operation again.

6.1. Message Type: Set Control Command 0x0201

Message Type: Set Control Command 0x0201

The host computer sends data to the radar: MSG_CFG_HUMAN_DETECTION_3D					
Format	Bytes	Basic types	Frame structure	Example frame	Frame meaning
SOF	1 byte	uint8	Starting frame	01	Used to set certain states of the radar.
ID	2 byte	uint16	Frame ID	00 00	
LEN	2 byte	uint16	Data frame length	00 04	
TYPE	2 byte	uint16	Frame type	02 01	
HEAD_CKSUM	1 byte	uint8	Header checksum	F3	
DATA	4 byte	int32	[command]	01 00 00 00	
DATA_CKSUM	1 byte	uint8	Data checksum	FE	

Used to set certain states of the radar:

- 0x1: Automatically generate interference zones.
- 0x2: Obtain the interference area and the detection area.
- 0x3: Clear the interference area.
- 0x4: Reset the detection area.
- 0x5: Get the hold delay time.
- 0x6: Enable point cloud display.
- 0x7: Disable point cloud display.
- 0x8: Open the target display.

- 0x9: Turn off the target display.
- 0xA: Set detection sensitivity: Low.
- 0xB: Set detection sensitivity: Medium.
- 0xC: Set detection sensitivity: High.
- 0xD: Get the detection sensitivity status.
- 0xE: Set trigger speed: Slow.
- 0xF: Set trigger speed: Medium.
- 0x10: Set trigger speed: Fast.
- 0x11: Get the trigger speed status.
- 0x12: Get the Z-axis range. **Note: This agreement applies only to 3D.**
- 0x13: Set installation method: top-mounted. **Note: This agreement applies only to 3D.**
- 0x14: Set installation method: side mounting. **Note: This agreement applies only to 3D.**
- 0x15: Get the installation method.
- 0x16: Set working mode: Low power consumption mode when no one is present.
- 0x17: Set working mode: Normal detection mode.
- 0x18: Get the operating mode.
- 0x19: Get the sleep time in low-power mode when no one is present.
- 0x1A: Reset to unmanned state.
- 0x1B: Set working mode: disable radar mode, P20 output high level.
- 0x1C: Set working mode: Disable radar mode, P20 output low level.
- 0x1D: Set P20 output: P20 outputs a high level when someone is present, and a low level when no one is present.
- 0x1E: Set P20 output: P20 outputs a low level when someone is present, and a high level when no one is present.
- 0x1F: Get the output status of P20 (indicating whether someone is present or not).
- 0x20: Set P20 output: P20 remains inactive and continuously outputs a low voltage level.
- 0x21: Set P20 output: P20 remains inactive, continuously outputting a high level signal.
- 0x22: Set P20 output: P20 operates in momentary mode, and outputs a 500ms low-level signal when the presence/absence detection status changes.
- 0x23: Set P20 output: P20 operates in momentary mode, and outputs a 500ms high-level signal when the presence/absence detection status changes.
- 0x24: Set working mode: High-reflectivity scene mode.
- 0x25: Clear the designated area.

- 0x26: Get the target lifecycle of the staging area.
- 0x27: Get the serial port output interval.

6.2. Message type: Setting the coordinate positions of interference areas, detection areas, and dwell areas 0x0202

The message type is 0x0202, and it only supports unidirectional data transmission mode.

The host computer sends data to the radar: MSG_CFG_HUMAN_DETECTION_3D_AREA					
Format	Bytes	Basic types	Frame structure	Example frame	Frame meaning
SOF	1 byte	uint8	Starting frame	01	Set the interference area, detection area, and dwell area.
ID	2 byte	uint16	Frame ID	00 00	
LEN	2 byte	uint16	Data frame length	00 1C	
TYPE	2 byte	uint16	Frame type	02 02	
HEAD_CKSUM	1 byte	uint8	Header checksum	/	
DATA	4 byte	int32	[area_id]	/	
DATA	4 byte	float	[x_min]	/	
DATA	4 byte	float	[x_max]	/	
DATA	4 byte	float	[y_min]	/	
DATA	4 byte	float	[y_max]	/	
DATA	4 byte	float	[z_min]	/	
DATA	4 byte	float	[z_max]	/	
DATA_CKSUM	1 byte	uint8	Data checksum	/	

Setting the interference area, detection area, and dwell area:

- [area_id]: Sets the ID of the area. Data type is int32. 0 to 3 represent interference areas, 4 to 7 represent detection areas, and 8 to 11 represent loitering areas.
- [x_min]: Sets the minimum x-coordinate of the region. Data type: float, unit: meters (m).
- [x_max]: Sets the maximum value of the x-coordinate of the region. Data type: float, unit: meters (m).
- [y_min]: Sets the minimum y-coordinate of the region. Data type: float, unit: meters (m).
- [y_max]: Sets the maximum value of the y-coordinate for the region. Data type: float, unit: meters (m).
- [z_min]: Sets the minimum z-coordinate of the region. Data type: float, unit: meters (m).
- [z_max]: Sets the maximum value of the z-coordinate for the region. Data type: float, unit: meters (m).

- **Note: There are a total of 4 interference zones, 4 detection zones, and 4 dwell zones. ID0 to ID3 are interference zones, ID4 to ID7 are detection zones, and ID8 to ID11 are dwell zones. Only one zone can be set at a time.**
- **Note: When using this agreement in a 2D context, please set z_min to -6m and z_max to 6m.**

6.3. Message type: Set hold delay time 0x0203

The message type is 0x0203, and it only supports unidirectional data transmission mode.

The host computer sends data to the radar: MSG_CFG_HUMAN_DETECTION_3D_PWM_DELAY					
Format	Bytes	Basic types	Frame structure	Example frame	Frame meaning
SOF	1 byte	uint8	Starting frame	01	Used to set the hold delay time.
ID	2 byte	uint16	Frame ID	00 00	
LEN	2 byte	uint16	Data frame length	00 04	
TYPE	2 byte	uint16	Frame type	02 03	
HEAD_CKSUM	1 byte	uint8	Header checksum	/	
DATA	4 byte	uint32	[pwm_delay]	1E	
DATA_CKSUM	1 byte	uint8	Data checksum	FE	

The following describes the effects of different values for DATA:

- [pwm_delay]: Sets the hold delay time. Data type: uint32, unit: seconds (s).
- Note: **The default value is 30 seconds.**

6.4. Message type: Set Z-axis range 0x0204

The message type is 0x0204, and it only supports unidirectional data transmission mode.

The host computer sends data to the radar: MSG_CFG_HUMAN_DETECTION_3D_Z					
Format	Bytes	Basic types	Frame structure	Example frame	Frame meaning
SOF	1 byte	uint8	Starting frame	01	Used to set the Z-axis range.
ID	2 byte	uint16	Frame ID	00 00	
LEN	2 byte	uint16	Data frame length	00 08	
TYPE	2 byte	uint16	Frame type	02 04	
HEAD_CKSUM	1 byte	uint8	Header checksum	/	
DATA	4 byte	float	[z_min]	/	
DATA	4 byte	float	[z_max]	/	
DATA_CKSUM	1 byte	uint8	Data checksum	FE	

The following describes the effects of different values for DATA:

- [z_min]: Sets the minimum z-coordinate of the region. Data type: float, unit: meters (m).
- [z_max]: Sets the maximum value of the z-coordinate for the region. Data type: float, unit: meters (m).
- **Note: This agreement applies only to 3D.**

6.5. Message type: Set low-power mode sleep time when no one is present (0x0205)

Message type: Set low-power mode sleep time when no one is present (0x0205)

Host computer sends data to radar: MSG_CFG_HUMAN_DETECTION_3D_LOW_POWER_MODE_TIME					
Format	Bytes	Basic types	Frame structure	Example frame	Frame meaning
SOF	1 byte	uint8	Starting frame	01	Used to set the hold delay time.
ID	2 byte	uint16	Frame ID	00 00	
LEN	2 byte	uint16	Data frame length	00 04	
TYPE	2 byte	uint16	Frame type	02 05	
HEAD_CKSUM	1 byte	uint8	Header checksum	/	
DATA	4 byte	unt32	[waitingPeriod]	01 F4	
DATA_CKSUM	1 byte	uint8	Data checksum	FE	

The following describes the effects of different values for DATA:

- [waitingPeriod]: Sets the sleep time in low-power mode when idle. Data type: uint32, unit: milliseconds (ms).
- Note: **The default value is 500ms.**

6.6. Message Type: Set dwell zone target lifecycle 0x0206

The message type is 0x0206, and it only supports unidirectional data transmission mode.

The host computer sends data to the radar: MSG_CFG_HUMAN_DETECTION_3D_STAY_LIFE					
Format	Bytes	Basic types	Frame structure	Example frame	Frame meaning
SOF	1 byte	uint8	Starting frame	01	Used to set the target lifespan of the dwell zone.
ID	2 byte	uint16	Frame ID	00 00	
LEN	2 byte	uint16	Data frame length	00 04	
TYPE	2 byte	uint16	Frame type	02 06	
HEAD_CKSUM	1 byte	uint8	Header checksum	/	
DATA	4 byte	uint32	[stayLife]	20	
DATA_CKSUM	1 byte	uint8	Data checksum	/	

The following describes the effects of different values for DATA:

- [stay_life]: Sets the target lifespan of the dwelling area. Data type: uint32. Decreases by 20 every second (s).
- Note: **The default value is 300, and it decreases by 20 every second (s).**

6.7. Message type: Set serial port output interval 0x0207

The message type is 0x0207, and it only supports unidirectional data transmission mode.

The host computer sends data to radar: MSG_CFG_HUMAN_DETECTION_3D_OUTPUT_INTERVAL					
Format	Bytes	Basic types	Frame structure	Example frame	Frame meaning
SOF	1 byte	uint8	Starting frame	01	Used to set the serial port output interval.
ID	2 byte	uint16	Frame ID	00 00	
LEN	2 byte	uint16	Data frame length	00 04	
TYPE	2 byte	uint16	Frame type	02 07	
HEAD_CKSUM	1 byte	uint8	Header checksum	/	
DATA	4 byte	uint32	[outputInterval]	1	
DATA_CKSUM	1 byte	uint8	Data checksum	/	

The following describes the effects of different values for DATA:

- [outputInterval]: Sets the serial port output interval; data type is uint32.
- **Note: The default value is 1, meaning output is generated for every frame. For example, setting it to 10 will result in output every 10 frames.**

6.8. Message type: Reporting personnel location 0x0A04 / 0x0A08

The message type is 0x0A04, which only supports unidirectional data transmission mode. 0x0A04 represents target data, and 0x0A08 represents point cloud data.

The radar sends data to the host computer: MSG_IND_HUMAN_DETECTION_3D_TGT_RES					
Format	Bytes	Basic types	Frame structure	Example frame	Frame meaning
SOF	1 byte	uint8	Starting frame	01	Used for reporting personnel location.
ID	2 byte	uint16	Frame ID	00 00	
LEN	2 byte	uint16	Data frame length	/	
TYPE	2 byte	uint16	Frame type	0A 04	
HEAD_CKSUM	1 byte	uint8	Header checksum	/	
DATA	4 byte	int32	[target_num]	/	
DATA	4 byte	float	[x]	/	
DATA	4 byte	float	[y]	/	
DATA	4 byte	float	[z]	/	
DATA	4 byte	int32	[dop_idx]	/	
DATA	4 byte	Int32	[cluster_id]	/	
…	…	…	…	…	
DATA_CKSUM	1 byte	uint8	Data checksum	/	

The following explains the meaning of each DATA bit:

- [target_num]: The number of targets.
- [x]: x-coordinate, data type float, unit: meters (m).
- [y]: y-coordinate, data type: float, unit: meters (m).
- [z]: z-coordinate, data type: float, unit: meters (m).
- [dop_idx]: Data type int32, speed dop_idx.
- [cluster_id]: Data type int32, the ID of the clustering target.

Note: When there are N targets, x, y, z, dop_idx, and cluster_id will also have N values.

- **Note: In 2D mode, the Z-axis output of this protocol is 0.**

6.9. Message type: Whether there are people in the reported area 0x0A0A

The message type is 0x0A0A, and it only supports unidirectional data transmission mode.

The radar sends data to the host computer: MSG_IND_HUMAN_DETECTION_3D_RES					
Format	Bytes	Basic types	Frame structure	Example frame	Frame meaning
SOF	1 byte	uint8	Starting frame	01	Are there any people in the reported area?
ID	2 byte	uint16	Frame ID	00 00	
LEN	2 byte	uint16	Data frame length	00 10	
TYPE	2 byte	uint16	Frame type	0A 0A	
HEAD_CKSUM	1 byte	uint8	Header checksum	F3	
DATA	4 byte	uint32	[detection_state_area0]	01	
DATA	4 byte	uint32	[detection_state_area1]	01	
DATA	4 byte	uint32	[detection_state_area2]	00	
DATA	4 byte	uint32	[detection_state_area3]	01	
DATA_CKSUM	1 byte	uint8	Data checksum	FE	

The following explains the meaning of each DATA bit:

- Report whether there are people in the four detection areas: 1 indicates someone is present, 0 indicates no one is present.

6.10. Message type: Reporting the locations of interference areas, detection areas, and dwelling areas (0x0A0B / 0x0A0C / 0x0A16)

Message type 0x0A0B indicates an interference zone message, 0x0A0C indicates a detection zone, and 0x0A16 indicates a dwell zone. Only unidirectional data transmission mode is supported.

The radar sends data to host computer: MSG_IND_HUMAN_(NOISE)DETECTION_3D_NOISE_AREA_RES					
Format	Bytes	Basic types	Frame structure	Example frame	Frame meaning
SOF	1 byte	uint8	Starting frame	01	When the host computer sends a command with data type 0x0201 and data value 0x02, the slave device will send three TF frames: 0x0A0B, 0x0A0C, and 0x0A16. 0x0A0B represents the interference area message, 0x0A0C represents the detection area, and 0x0A16 represents the dwell area.
ID	2 byte	uint16	Frame ID	00 00	
LEN	2 byte	uint16	Data frame length	/	
TYPE	2 byte	uint16	Frame type	0A 0B/0A 0C	
HEAD_CKSUM	1 byte	uint8	Header checksum	/	
DATA	4 byte	int32	[x_min]	/	
DATA	4 byte	float	[x_max]	/	
DATA	4 byte	float	[y_min]	/	
DATA	4 byte	float	[y_max]	/	
DATA	4 byte	float	[z_min]	/	
DATA	4 byte	float	[z_max]	/	
…	…	…	…	…	
DATA_CKSUM	1 byte	uint8	Data checksum	FE	

The following explains the meaning of each DATA bit:

- [x_min]: Sets the minimum x-coordinate of the region. Data type: float, unit: meters (m).
- [x_max]: Sets the maximum value of the x-coordinate of the region. Data type: float, unit: meters (m).
- [y_min]: Sets the minimum y-coordinate of the region. Data type: float, unit: meters (m).
- [y_max]: Sets the maximum value of the y-coordinate for the region. Data type: float, unit: meters (m).
- [z_min]: Sets the minimum z-coordinate of the region. Data type: float, unit: meters (m).
- [z_max]: Sets the maximum value of the z-coordinate for the region. Data type: float, unit: meters (m).

Note: This message reports the locations of the interference areas, detection areas, and dwell areas. There are a total of 4 interference areas, 4 detection areas, and 4 dwell areas. 0x0A0B reports the coordinates of the 4 interference areas, 0x0A0C reports the coordinates of the 4 detection areas, and 0x0A16 reports the coordinates of the 4 dwell areas.

- **Note: This agreement applies only to 2D scenarios, where z_min is defaulted to -6m and z_max is defaulted to 6m.**

6.11. Message type: Report keep-alive delay time 0x0A0D

The message type is 0x0A0D, and it only supports unidirectional data transmission mode.

The radar sends data to the host computer: MSG_IND_HUMAN_DETECTION_3D_PWM_DELAY					
Format	Bytes	Basic types	Frame structure	Example frame	Frame meaning
SOF	1 byte	uint8	Starting frame	01	Get the hold delay time.
ID	2 byte	uint16	Frame ID	00 01	
LEN	2 byte	uint16	Data frame length	00 04	
TYPE	2 byte	uint16	Frame type	0A 0D	
HEAD_CKSUM	1 byte	uint8	Header checksum	FC	
DATA	4 byte	uint32	[pwmDelayTimer]	05	
DATA_CKSUM	1 byte	uint8	Data checksum	FA	

The following explains the meaning of each DATA bit:

- Maintain the delay time.

6.12. Message Type: Report Detection Sensitivity Status 0x0A0E

The message type is 0x0A0E, and it only supports unidirectional data transmission mode.

The radar sends data to the host computer: MSG_IND_HUMAN_DETECTION_3D_DETECT_SENSITIVITY					
Format	Bytes	Basic types	Frame structure	Example frame	Frame meaning
SOF	1 byte	uint8	Starting frame	01	Get the detection sensitivity status.
ID	2 byte	uint16	Frame ID	00 01	
LEN	2 byte	uint16	Data frame length	00 01	
TYPE	2 byte	uint16	Frame type	0A 0E	
HEAD_CKSUM	1 byte	uint8	Header checksum	FF	
DATA	1 byte	uint8	[detectSensitivity]	01	
DATA_CKSUM	1 byte	uint8	Data checksum	FE	

The following explains the meaning of each DATA bit:

- Reported detection sensitivity: 0: Low, 1: Medium, 2: High.

6.13. Message type: Report triggered speed status 0x0A0F

The message type is 0x0A0F, and it only supports unidirectional data transmission mode.

The radar sends data to the host computer: MSG_IND_HUMAN_DETECTION_3D_DETECT_TRIGGER					
Format	Bytes	Basic types	Frame structure	Example frame	Frame meaning
SOF	1 byte	uint8	Starting frame	01	Get the trigger speed status.
ID	2 byte	uint16	Frame ID	00 01	
LEN	2 byte	uint16	Data frame length	00 01	
TYPE	2 byte	uint16	Frame type	0A 0F	
HEAD_CKSUM	1 byte	uint8	Header checksum	FB	
DATA	1 byte	uint8	[detectTrigger]	02	
DATA_CKSUM	1 byte	uint8	Data checksum	FD	

The following explains the meaning of each DATA bit:

- Reported trigger speed: 0: Slow, 1: Medium, 2: Fast.

6.14. Message type: Report Z-axis range 0x0A10

The message type is 0x0A10, and it only supports unidirectional data transmission mode.

The radar sends data to the host computer: MSG_IND_HUMAN_DETECTION_3D_DETECT_TRIGGER					
Format	Bytes	Basic types	Frame structure	Example frame	Frame meaning
SOF	1 byte	uint8	Starting frame	01	Get the Z-axis setting range.
ID	2 byte	uint16	Frame ID	00 01	
LEN	2 byte	uint16	Data frame length	00 08	
TYPE	2 byte	uint16	Frame type	0A 10	
HEAD_CKSUM	1 byte	uint8	Header checksum	FB	
DATA	4 byte	float	[z_min]	/	
DATA	4 byte	float	[z_max]	/	
DATA_CKSUM	1 byte	uint8	Data checksum	FD	

The following explains the meaning of each DATA bit:

- [z_min]: Sets the minimum z-coordinate of the region. Data type: float, unit: meters (m).
- [z_max]: Sets the maximum value of the z-coordinate for the region. Data type: float, unit: meters (m).
- **Note: This agreement applies only to 3D.**

6.15. Message type: Report installation method 0x0A11

The message type is 0x0A11, and it only supports unidirectional data transmission mode.

The radar sends data to the host computer: MSG_IND_HUMAN_DETECTION_3D_INSTALL_SITE					
Format	Bytes	Basic types	Frame structure	Example frame	Frame meaning
SOF	1 byte	uint8	Starting frame	01	Get installation instructions.
ID	2 byte	uint16	Frame ID	02 2E	
LEN	2 byte	uint16	Data frame length	00 01	
TYPE	2 byte	uint16	Frame type	0A 11	
HEAD_CKSUM	1 byte	uint8	Header checksum	C8	
DATA	1 byte	uint8	[installSite]	/	
DATA_CKSUM	1 byte	uint8	Data checksum	FD	

The following explains the meaning of each DATA bit:

- Report the installation method: 0: top mounting, 1: side mounting.

6.16. Message type: Reporting operating mode 0x0A12

The message type is 0x0A12, and it only supports unidirectional data transmission mode.

The radar sends data to the host computer: MSG_IND_HUMAN_DETECTION_3D_LOW_POWER_MODE					
Format	Bytes	Basic types	Frame structure	Example frame	Frame meaning
SOF	1 byte	uint8	Starting frame	01	Get the operating mode.
ID	2 byte	uint16	Frame ID	02 2E	
LEN	2 byte	uint16	Data frame length	00 01	
TYPE	2 byte	uint16	Frame type	0A 12	
HEAD_CKSUM	1 byte	uint8	Header checksum	/	
DATA	1 byte	uint32	[lowPowerMode]	/	
DATA_CKSUM	1 byte	uint8	Data checksum	FD	

The following explains the meaning of each DATA bit:

- 0: Normal detection mode.
- 1. Low power consumption mode when unoccupied.
- 2. Turn off radar mode; P20 outputs a high voltage level.
- 3. Turn off radar mode; P20 outputs a low signal.
- 4. High-reflectivity scene mode.

6.17. Message Type: Report of low-power mode sleep time when unmanned (0x0A13)

The message type is 0x0A12, and it only supports unidirectional data transmission mode.

The radar sends data to the host computer: MSG_IND_HUMAN_DETECTION_3D_LOW_POWER_TIME					
Format	Bytes	Basic types	Frame structure	Example frame	Frame meaning
SOF	1 byte	uint8	Starting frame	01	Get the sleep time in low-power mode when no one is present.
ID	2 byte	uint16	Frame ID	02 2E	
LEN	2 byte	uint16	Data frame length	00 01	
TYPE	2 byte	uint16	Frame type	0A 13	
HEAD_CKSUM	1 byte	uint8	Header checksum	/	
DATA	4 byte	uint32	[waitingPeriod]	/	
DATA_CKSUM	1 byte	uint8	Data checksum	FD	

The following explains the meaning of each DATA bit:

- Sleep time in low-power mode when no one is present.

6.18. Message type: Report of unmanned low-power mode operating status 0x0A14

The message type is 0x0A14, and it only supports unidirectional data transmission mode.

The radar sends data to the host computer: MSG_IND_HUMAN_DETECTION_3D_MODE					
Format	Bytes	Basic types	Frame structure	Example frame	Frame meaning
SOF	1 byte	uint8	Starting frame	01	Report the unmanned system's low-power operating status.
ID	2 byte	uint16	Frame ID	02 2E	
LEN	2 byte	uint16	Data frame length	00 01	
TYPE	2 byte	uint16	Frame type	0A 14	
HEAD_CKSUM	1 byte	uint8	Header checksum	/	
DATA	1 byte	uint8	[detectStateMessager]	/	
DATA_CKSUM	1 byte	uint8	Data checksum	FD	

The following explains the meaning of each DATA bit:

- 0: The operating status is unmanned, low-power mode.
- 1. The device is operating in normal mode.

Note: In unattended low-power mode, the device will enter low-power mode when no one is present and switch to normal mode when someone is detected. This message will only be sent when the mode is set to unattended low-power mode, and it will be sent when switching between unattended low-power mode and normal mode.

6.19. Message type: Report GPIO status change 0x0A15

The message type is 0x0A15, and it only supports unidirectional data transmission mode.

The radar sends data to the host computer: MSG_IND_HUMAN_DETECTION_3D_GPIO_SWITCH					
Format	Bytes	Basic types	Frame structure	Example frame	Frame meaning
SOF	1 byte	uint8	Starting frame	01	Report GPIO state changes.
ID	2 byte	uint16	Frame ID	02 2E	
LEN	2 byte	uint16	Data frame length	00 01	
TYPE	2 byte	uint16	Frame type	0A 15	
HEAD_CKSUM	1 byte	uint8	Header checksum	/	
DATA	1 byte	uint8	[gpioOutputSwitch]	/	
DATA_CKSUM	1 byte	uint8	Data checksum	FD	

The following explains the meaning of each DATA bit:

- 0: When someone is present, P20 outputs a high level; when no one is present, P20 outputs a low level.
- 1. When someone is present, P20 outputs a low signal; when no one is present, P20 outputs a high signal.
- 2. P20 remains inactive, continuously outputting a low voltage level.
- 3: P20 remains inactive, continuously outputting a high logic level.
- 4. P20 operates in jog mode, and switches between occupied and unoccupied states, outputting a 500ms low-level signal.
- 5: P20 operates in jog mode, and outputs a 500ms high-level signal upon switching between occupied and unoccupied states.

6.20. Message Type: Reporting the lifecycle of a target in the designated area (0x0A17)

The message type is 0x0A17, and it only supports unidirectional data transmission mode.

The radar sends data to the host computer: MSG_IND_HUMAN_DETECTION_3D_STAY_LIFE					
Format	Bytes	Basic types	Frame structure	Example frame	Frame meaning
SOF	1 byte	uint8	Starting frame	01	Get the target lifecycle of the staging area.
ID	2 byte	uint16	Frame ID	00 01	
LEN	2 byte	uint16	Data frame length	00 04	
TYPE	2 byte	uint16	Frame type	0A 17	
HEAD_CKSUM	1 byte	uint8	Header checksum	/	
DATA	4 byte	uint32	[stayLife]	1	
DATA_CKSUM	1 byte	uint8	Data checksum	/	

The following explains the meaning of each DATA bit:

- The target lifecycle of the staging area.

6.21. Message type: Report serial port output interval 0x0A18

The message type is 0x0A18, and it only supports unidirectional data transmission mode.

The radar sends data to the host computer: MSG_IND_HUMAN_DETECTION_3D_OUTPUT_INTERVAL					
Format	Bytes	Basic types	Frame structure	Example frame	Frame meaning
SOF	1 byte	uint8	Starting frame	01	Get the serial port output interval.
ID	2 byte	uint16	Frame ID	00 01	
LEN	2 byte	uint16	Data frame length	00 04	
TYPE	2 byte	uint16	Frame type	0A 18	
HEAD_CKSUM	1 byte	uint8	Header checksum	/	
DATA	4 byte	uint32	[outputInterval]	1	
DATA_CKSUM	1 byte	uint8	Data checksum	/	

The following explains the meaning of each DATA bit:

- Serial port output interval.

6.22. Message type: Set Bluetooth serial port baud rate 0x0F0F

The message type is 0x0F0F, and it only supports unidirectional data transmission mode.

The host computer sends data to the Bluetooth module: MSG_CFG_HUMAN_DETECTION_3D_BAUD					
Format	Bytes	Basic types	Frame structure	Example frame	Frame meaning
SOF	1 byte	uint8	Starting frame	01	Used to set the serial port baud rate.
ID	2 byte	uint16	Frame ID	00 00	
LEN	2 byte	uint16	Data frame length	00 01	
TYPE	2 byte	uint16	Frame type	0F 0F	
HEAD_CKSUM	1 byte	uint8	Header checksum	/	
DATA	1 byte	Uint8	[baud index]	1	
DATA_CKSUM	1 byte	uint8	Data checksum	/	

The following describes the effects of different values for DATA:

- [baud index]: Sets the serial port baud rate; data type is uint8.
- **Note: The default values are 1 and 115200.**
-

Baud index	Baud
0	9600
1	115200
2	128300
3	256000
4	460800
5	921600

Example: 01 00 00 00 01 0f 0f xx 00 00

7. Programming interface

7.1. Encoding TF messages

void tinyFramefTx(TF_TYPE type, uint8 *data, TF_LEN len);

`type` specifies the data type being sent, of type uint16. For example, for reporting person detection results, the data type is 0x0A10. See section 4.2.1.6 for details.

`Uint8* data` is the address of the data to be sent.

`Len` is the length of the data being sent, of type uint16.

7.2. Decoding TF messages

TinyFrameRx tinyFramefRx(void);

After successfully receiving the message, the received data is returned to a variable of type TinyFrameRx.

A. The following is the method for extracting data from the DATA field:

For example, the serial port receives hexadecimal data, and converts it from int32 to float: If the [x_point] field is 0x66, 0x66, 0xA2, 0x41, first combine them into a uint32 integer. Since the TF frame Data field uses little-endian order, the value is 0x41A26666. Then, perform a type cast to float. The final result is 20.3.

```
int main(void)
{
    unsigned int param = 0x41A26666;
    float res = *(float *)&param;

    printf("data: %f\n", res);
    return 0;
}
```

B. The following is an explanation of each CKSUM:

HEAD_CKSUM: TF frame header checksum [from the first byte to the byte before the HEAD_CKSUM field]

DATA_CKSUM: TF data checksum [from the first byte of DATA to the byte before the DATA_CKSUM field]

The C code for calculating the CKSUM is shown below:

```
unsigned char getCksum(unsigned char *data, unsigned char len)
{
  unsigned char ret = 0;

  for (int i = 0; i < len; i++)
    ret = ret ^ data[i];

  ret = ~ret;

  return ret;
}

```

7.3. Example code

If you would like a demo for parsing TF frame data (including C language demos for Linux and Keil µVision5 environments, and a Python language demo), please contact our sales team.

8. Contact Information

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